

Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - I

Course Code : CSE 0533 1101 (CSE) & PHY 0533 1101 (EEE)

Term: Spring – 2024	Date: 29.01.2024	Time: 10.00 AM	Room: 327	Activity: Lecture 9
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Topics to be discussed: Capacitance, calculation of capacitance, different types of capacitors, capacitors with dielectric, energy storage in an electric field.

Target CLO & PLO : CLO1,2,3 & PLO 2 & 5

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define Capacitance and dielectric constant
- ❖ Describe different types of capacitor and uses
- ❖ Derive the expression for the energy of a charged capacitor
- ❖ Deduce the expression for the capacitance of combination of capacitors
- ❖ Solve some recommended mathematical problems

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion on some natural phenomenon	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

- 1) Capacitance depends on -
a) Size b) charge c) voltage d) emf e) magnetic field
- 2) $1 F = ?$
a) $1C/1A$ b) $1C/1V$ c) $1A/1m$ d) $1A/1C$ e) $1A/1V$

Sample Questions (SAQ):

- 1) Define capacitance and dielectric constant.
- 2) Give the construction of the Parallel plate capacitor. Calculate the capacitance of a parallel plate capacitor of area A and plate separation d.
- 3) A spherical capacitor consists of two concentric spherical shell of radii a and b, with $b > a$, show that the capacitance is $C = 4\pi\epsilon_0 \frac{ab}{b-a}$
- 4) Prove that, the capacitance of a cylindrical capacitor is independent of charge of the capacitor and the potential difference between the plates of the capacitor.
- 5) The parallel plates of an air-filled capacitor are everywhere 1.0 mm apart. What must be the plate area if the capacitance is to be 1.0 farad?
- 6) The parallel plate of a Ruby-Mica filled capacitor are every 2.0 mm apart. What must be the plate area be if the capacitance is to be 2.0 F? Given dielectric constant for Ruby-Mica is 1.5.
- 6) If three capacitors of $3\mu F$, $2\mu F$ and $5\mu F$ are connected in parallel across a voltage of 100V. Calculate the equivalent capacitance and stored energy.

Reference Books:

1. Giasuddin ahmed – Physics for Engineers (Part-2)
2. Huq. Rafiqullah- Electricity and Magnetism

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 359-368
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 37-86